Solution (i) $8x^4y^2,\ 48x^2y^4$

First let us find the LCM of the numerical coefficients.

That is, LCM $(8,48) = 2\times2\times2\times6 = 48$

Then find the LCM of the terms involving variables.

That is, LCM $(x^4y^2, x^2y^4) = x^4y^4$

Finally find the LCM of the given expression.

We condclude that the LCM of the given expression is the product of the LCM of the numerical coefficient and the LCM of the terms with variables.

Therefore, LCM $(8x^4y^2, 48x^2y^4) = 48x^4y^4$

(ii) $(5x-10), (5x^2-20)$

$$5x-10 = 5(x-2)$$

$$5x^2-20 = 5(x^2-4) = 5(x+2)(x-2)$$

Therefore, LCM $[(5x-10),(5x^2-20)] = 5(x+2)(x-2)$

(iii) $(x^4-1),\ x^2-2x+1$

$$x^4-1 = (x^2)^2-1 = (x^2+1)(x^2-1) = (x^2+1)(x+1)(x-1)$$

$$x^2-2x+1 = (x-1)^2$$

Therefore, LCM $[(x^4-1),(x^2-2x+1)] = (x^2+1)(x+1)(x-1)^2$

(iv) $x^3-27, (x-3)^2, x^2-9$

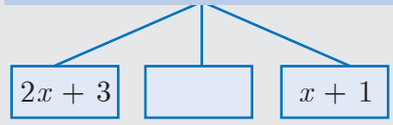
$x^3-27 = (x-3)(x^2+3x+9)$; $(x-3)^2 = (x-3)^2$; $(x^2-9) = (x+3)(x-3)$

Therefore, LCM $[(x^3-27),(x-3)^2,(x^2-9)] = (x-3)^2(x+3)(x^2+3x+9)$

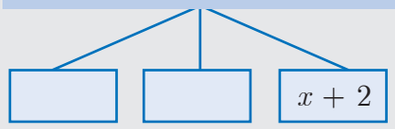
Thinking Corner

Complete the factor tree for the given polynomials $f(x)$ and $g(x)$. Hence find their GCD and LCM.

$f(x) = 2x^3 - 9x^2 - 32x - 21$



$g(x) = 2x^3 - 7x^2 - 43x - 42$



GCD $[f(x)$ and $g(x)]$ = _____ LCM $[f(x)$ and $g(x)]$ = _____

Exercise 3.2

1. Find the GCD of the given polynomials

(i) $x^4+3x^3-x-3,\ x^3+x^2-5x+3$ (ii) $x^4-1,\ x^3-11x^2+x-11$

(iii) $3x^4+6x^3-12x^2-24x,\ 4x^4+14x^3+8x^2-8x$

(iv) $3x^3+3x^2+3x+3,\ 6x^3+12x^2+6x+12$

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